

Lecture 5: Conductors

Pre-Lecture Worksheet

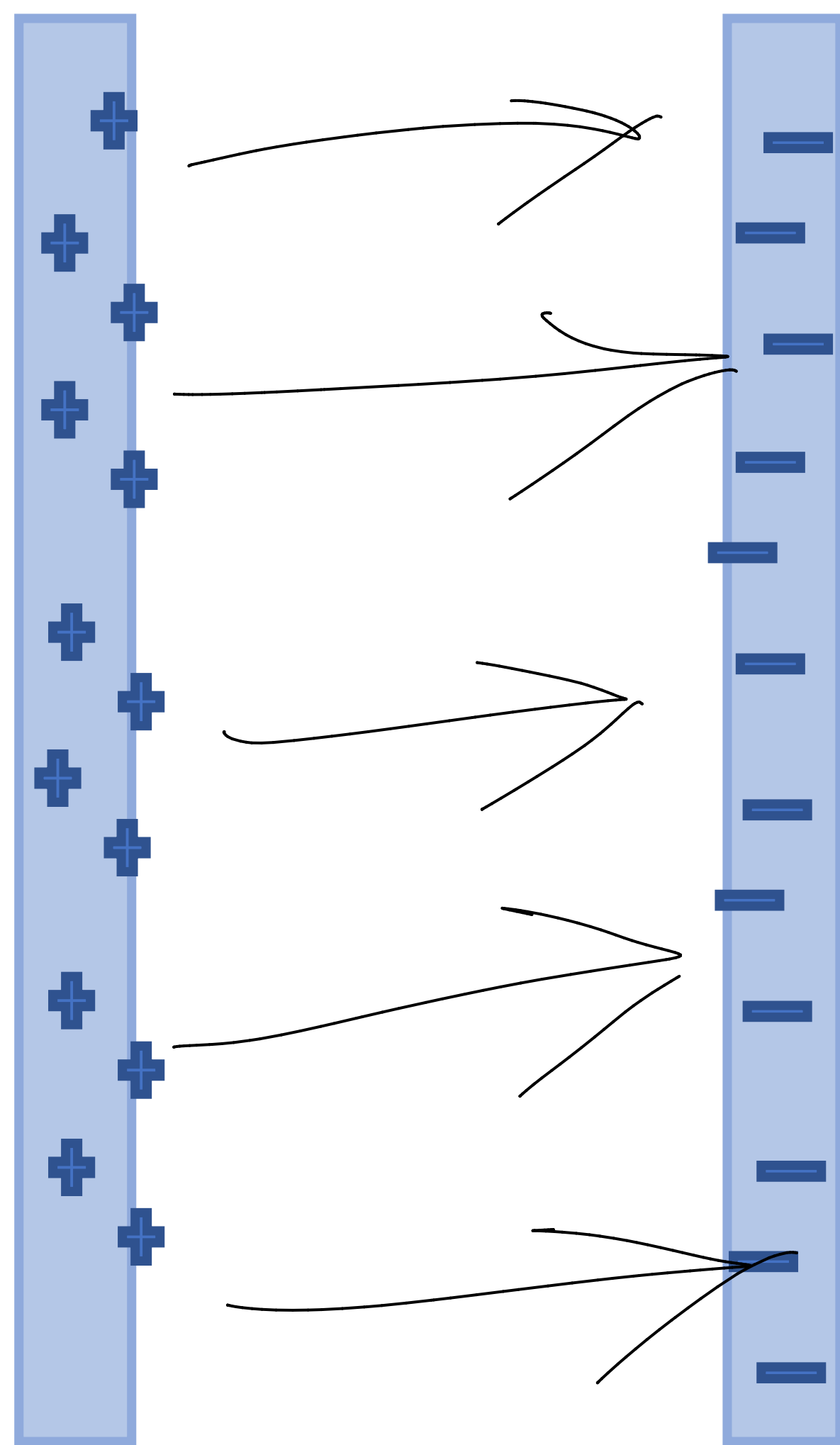
Complete the following worksheet in preparation for the next lecture. The following questions correspond to material covered in chapter 21. Feel free to read chapter 21 and answer the questions. You may also seek answers to the questions from other sources. Note: the textbook will be used as the final source for these questions. While answers gleaned from other sources may be correct, answers must encompass the material covered in the book to be considered correct.

Conductors are materials that can carry and electric charge. Answer the following questions about their general properties.

- 1) (True or False) The electric field inside a conductor is 0.
- 2) (True or False) The net charge of on a conductor distributes itself throughout the conductor.
- 3) The electric field created by the conductor has field lines that are (parallel or perpendicular) to the surface of the conductor.

4) Draw the field lines for the following parallel plate conductors.

A



B

